

Full Stack Web Development with JavaScript

Course Objective:

- 1) Identify the core technologies in full-stack JavaScript development.
- 2) Explain the architecture of web applications and the roles of frontend and backend.
- 3) Build interactive frontend interfaces using HTML, CSS, TailwindCSS, JavaScript, and React.
- 4) Develop server-side applications using Node.js, Express.js, and EJS.
- 5) Integrate MongoDB for backend data storage and implement RESTful APIs.
- 6) Deploy and optimize full-stack web applications for performance and security.

Learning Outcome:

After completing the course, the students should be able to:

- 1) Understand the principles of full-stack web development using the JavaScript ecosystem.
- 2) Build responsive UIs using HTML, CSS, TailwindCSS, and React.
- 3) Handle backend development using Node.js, Express.js, and EJS templating.
- 4) Implement RESTful APIs and work with MongoDB for data persistence.
- 5) Apply best practices for code structure, scalability, and web security.

Course Contents:

UNIT NUMBER	CONTENTS	NO. OF HOURS
MODULE-I	Frontend Development Basics • Web Development Overview and Client-Server Architecture. • HTML5 & CSS3 Essentials • Introduction to TailwindCSS: Utility-first styling • JavaScript Fundamentals: DOM, events, arrays, loops, functions • Building interactive interface using these technologies	6

MODULE-II	Server-Side Development with Node.js & Express.js • Introduction to Node.js: Asynchronous JS and event-driven architecture. • Working with NPM and creating Node.js apps. • Express.js fundamentals: Routing, Middleware, and Request Handling. • EJS Templating Engine: Server-rendered pages. • File handling and form data in Express	10
MODULE-III	Database and API Development: • Introduction to MongoDB: NoSQL database and data modeling • CRUD operations with MongoDB and Mongoose. • RESTful API design and integration. • User Authentication with JWT and sessions. • Role-based access and secure endpoints	10
MODULE-IV	Advanced Frontend with React.js: • Introduction to React: Components, Props, State. • Functional components and React Hooks (useState, useEffect). • Routing with React Router. • State management with Context API. • Integrating REST APIs into React.	10
MODULE-V	Full-Stack Project Development & Deployment: • Build a complete web application. • Connect frontend and backend using REST APIs. • Deploying the application using Heroku, Netlify, or Vercel. • Best practices: Code organization, error handling, and performance optimization.	8
MODULE-VI	Website Launch and SEO Optimization: • Launch strategy using social media, email, and paid ads. • Keyword research and on-page optimization. • Optimizing titles, meta tags, and content structure • Content planning for traffic growth. • Using Google Analytics and Search Console for performance tracking	6
PROJECTS	1. Weather application –Frontend 2. Food ordering website – Frontend 3. Product launch and SEO optimization 4. Social media website – Backend 5. E-commerce website - Frontend and Backend	10

Text/Reference Books:

1. *Learning JavaScript* by Ethan Brown
2. *Web Development with Node and Express* by Ethan Brown
3. *Tailwind CSS: From Zero to Production* by Adam Wathan
4. *Full-Stack React Projects* by Shama Hoque
5. *MongoDB: The Definitive Guide* by Shannon Bradshaw

Career Prospects

- Full Stack Developer (JavaScript).
- Frontend Developer (React.js).
- Backend Developer (Node.js, Express.js).
- API Developer (REST/MongoDB).
- Web Designer (TailwindCSS + HTML/CSS).